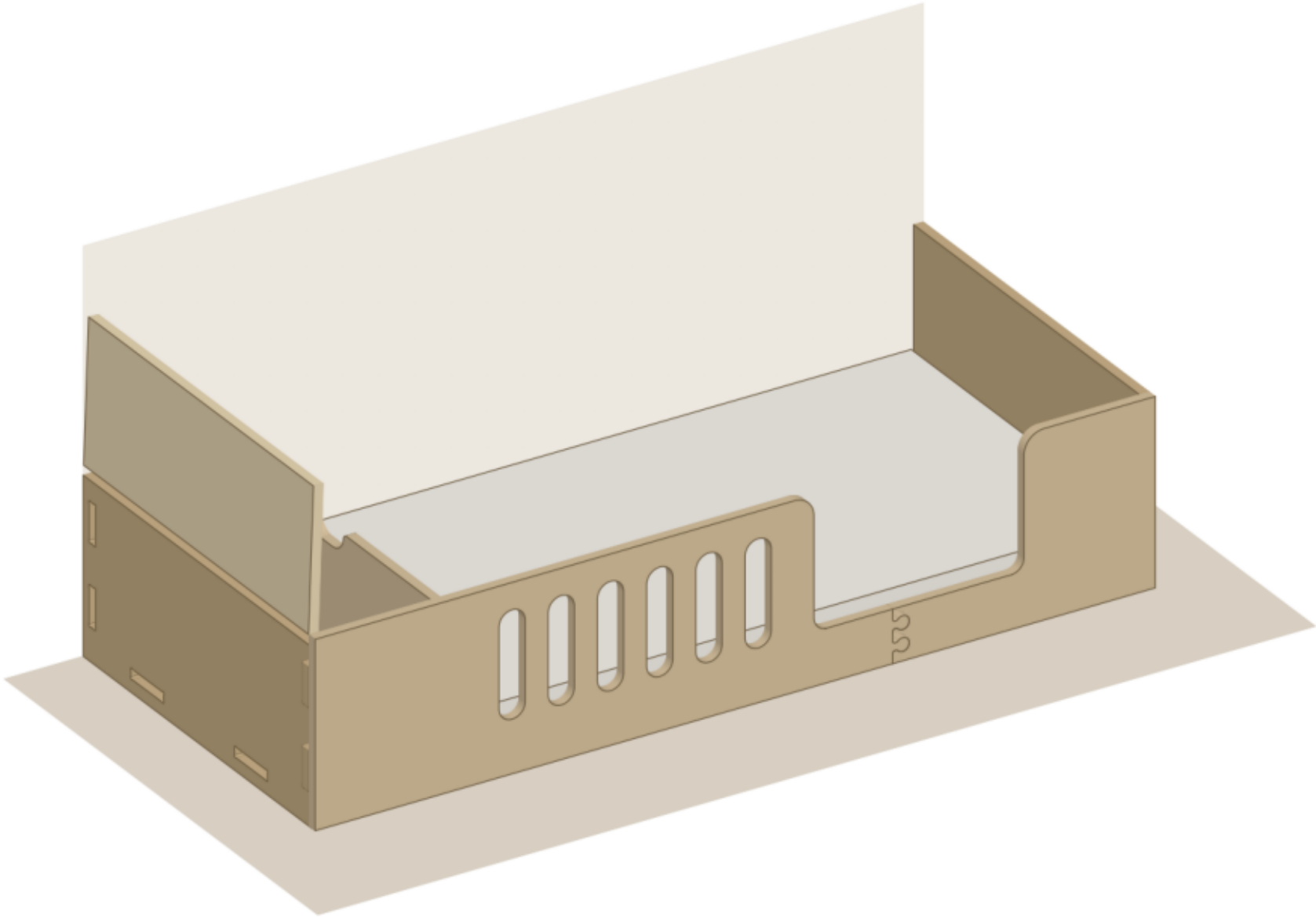
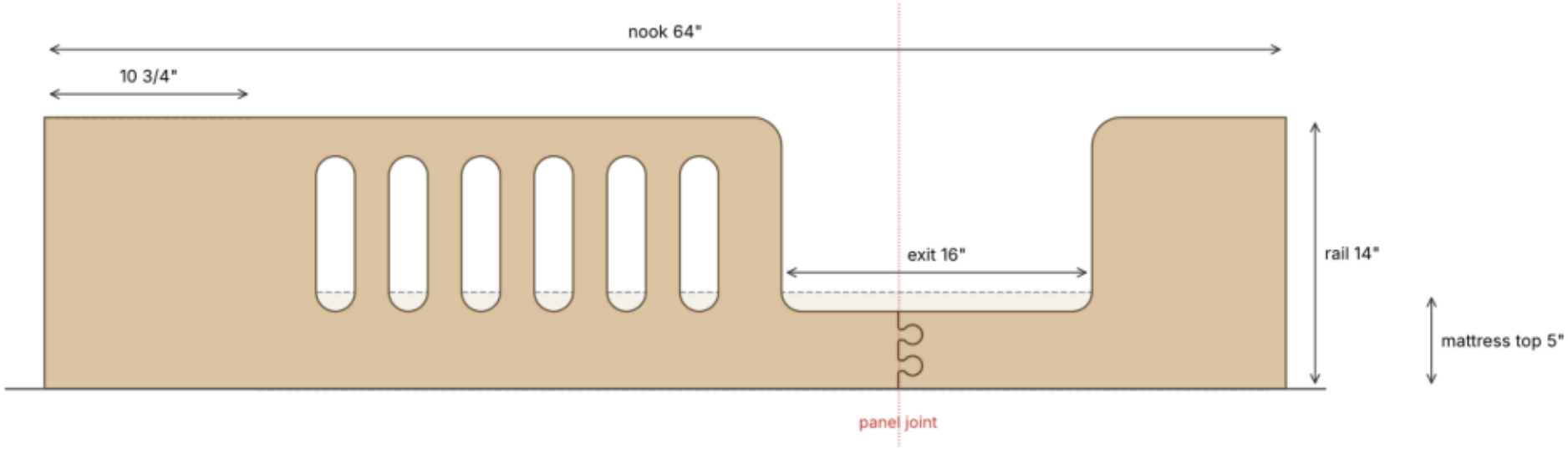
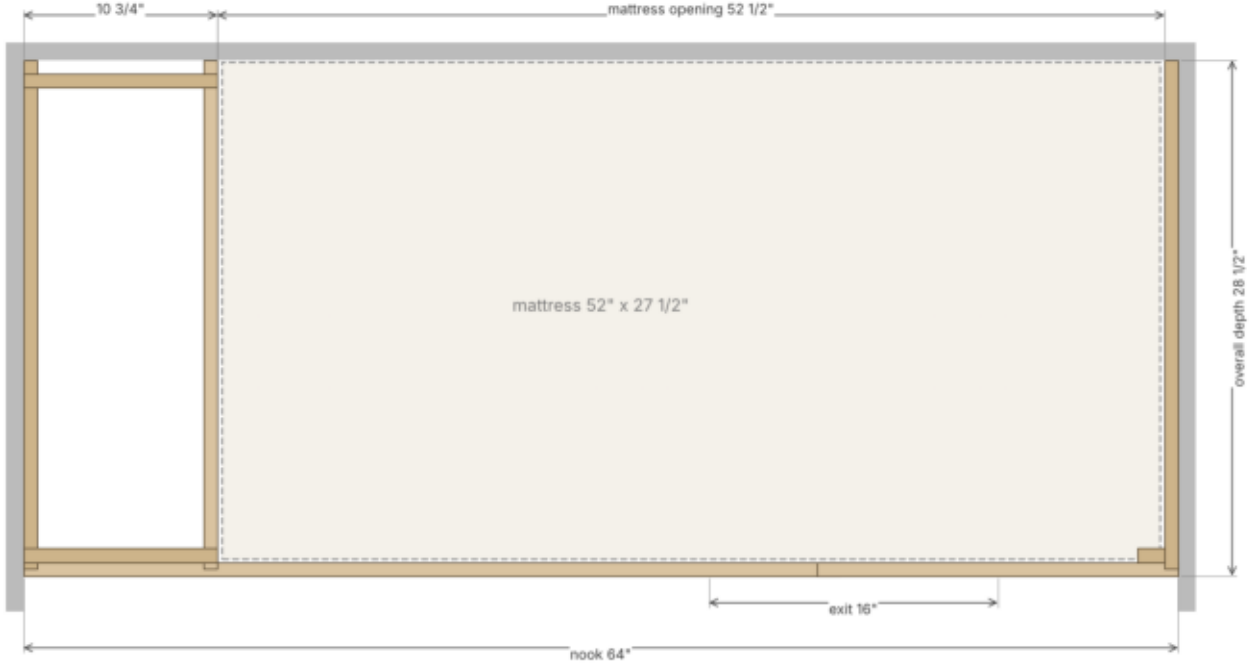
Floor bed, side exit, one box, flush tops



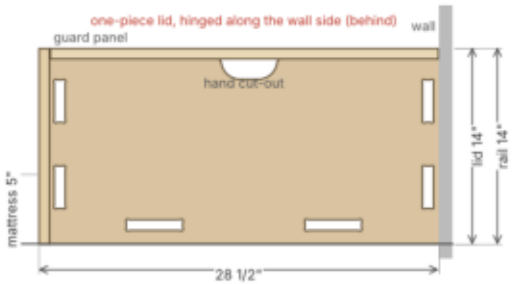
Floor bed, side exit, one box, flush tops: front view



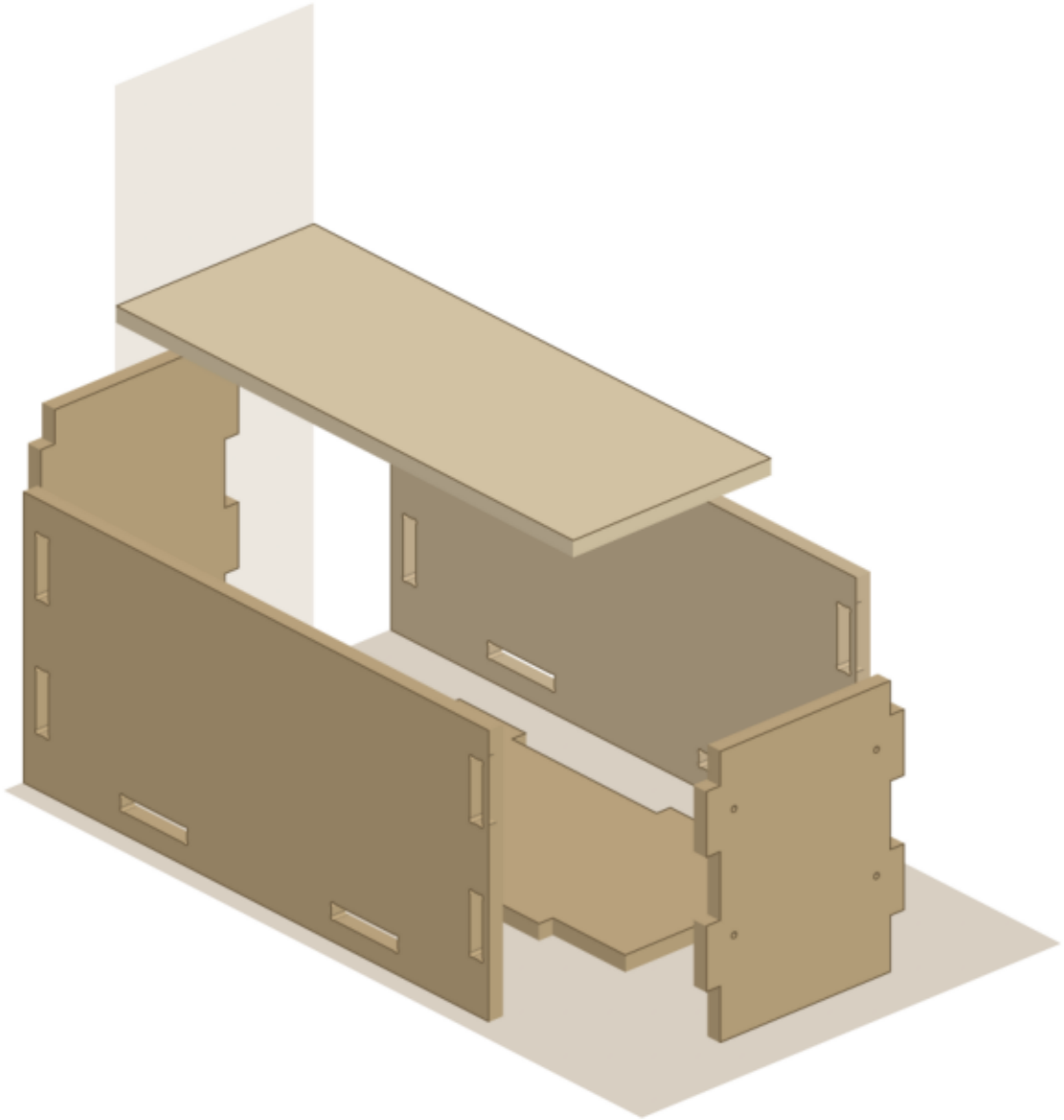
Plan (front of bed at the bottom)




Box seen from the mattress

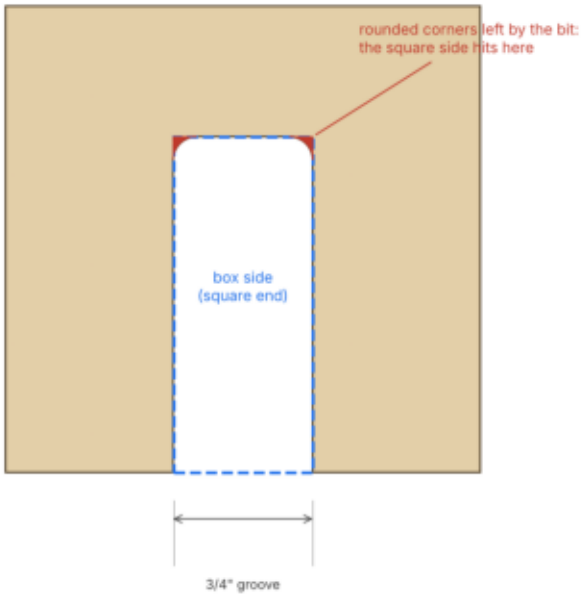


Head box, exploded (tabs pass through the sides)

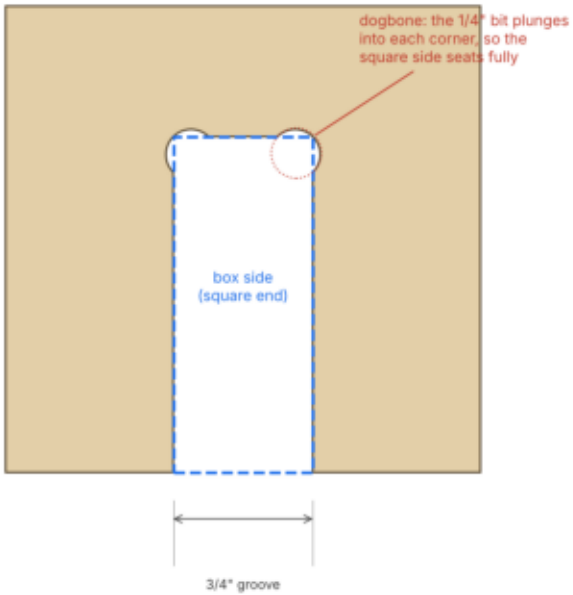


Dogbones: how a square part fits a CNC-cut groove (looking at the back of the front panel)

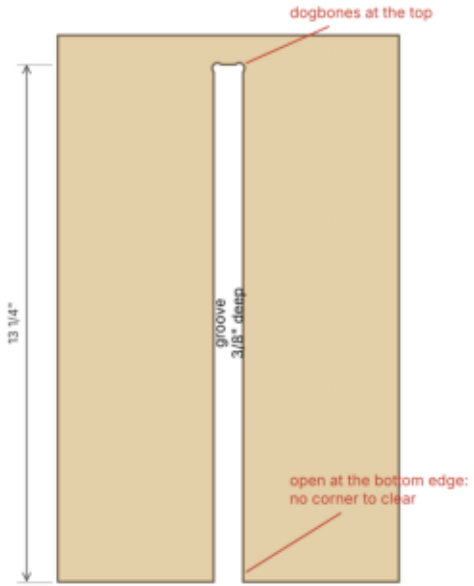
Without dogbones (red = wood that blocks the square corner)




With dogbones

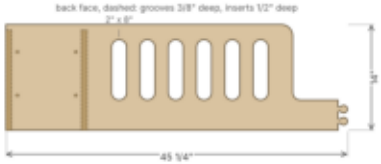



Back of the front panel, one groove

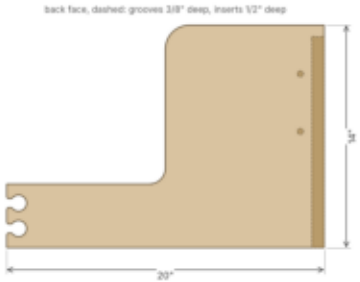


# Floor bed, side exit, one box, flush tops: part drawings (inches, 3/4" plywood)

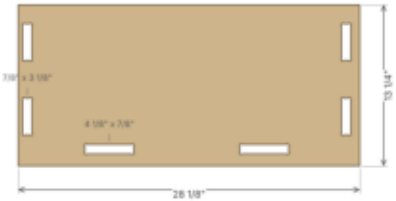
Front panel A  (qty 1)
knob half



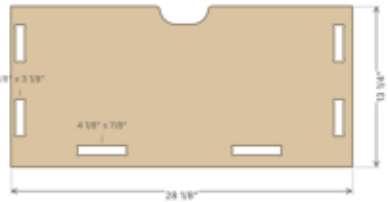

Front panel B  (qty 1)
socket half



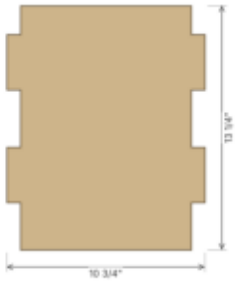

Box side (wall)  (qty 1)
against the nook wall, 1 per box



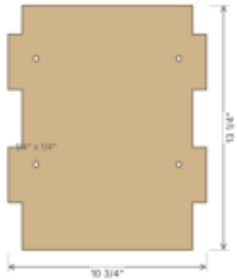

Box side (mattress)  (qty 1)
faces the mattress; hand cut-out to open the lid




Box back  (qty 1)
screw through into wall studs




Box front  (qty 1)
against the back of the front panel; 4 bolts into it



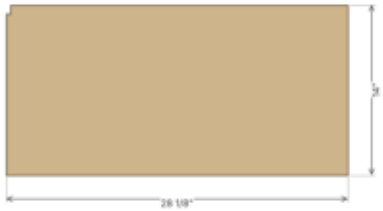

Box floor  (qty 1)
sits 1" off the floor




Box lid  (qty 1)
piano hinge along the long wall-side edge




Foot wall liner  (qty 1)
screw to the foot wall studs; runs into the panel groove

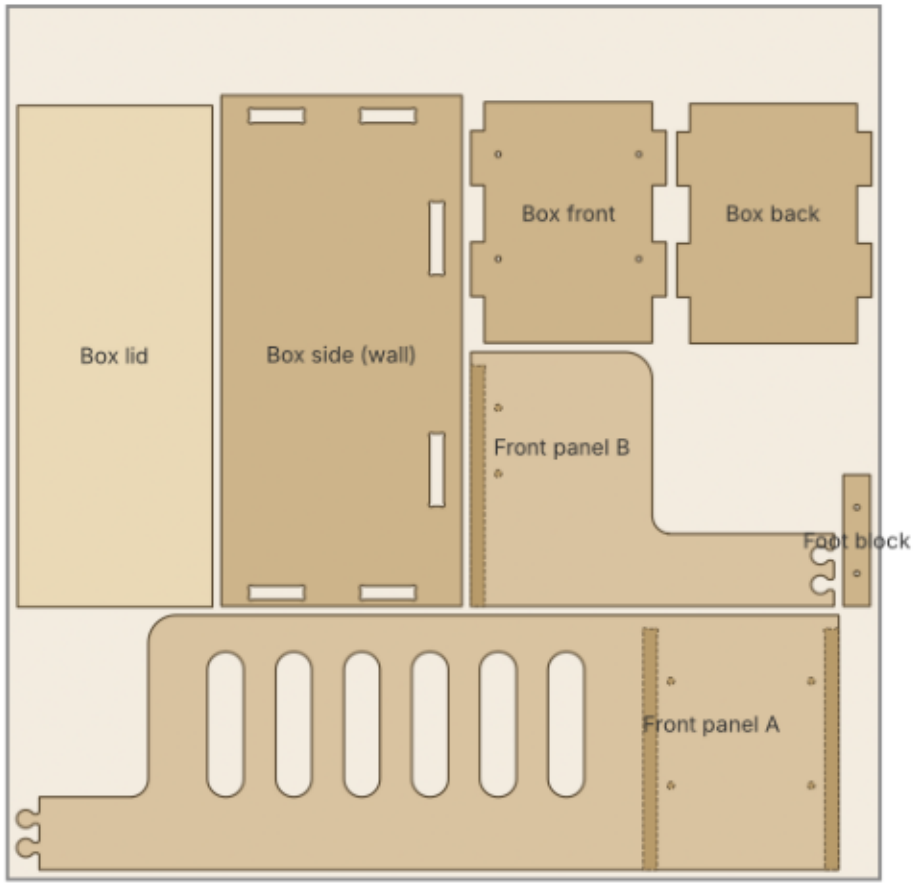



Foot block  (qty 1)
above the mattress; glue + 2 screws to the liner, 2 bolts into the panel



Floor bed, side exit, one box, flush tops: nesting on 48" square panels

4'x4' panel 1




4'x4' panel 2



# Floor bed, side exit, one box, flush tops: cut list, hardware, assembly

## Cut list

| Part | Qty | Size (W x H) | Notes |
|---|---|---|---|
| Front panel A | 1 | 45 1/4" x 14" | knob half |
| Front panel B | 1 | 20" x 14" | socket half |
| Box side (wall) | 1 | 28 1/8" x 13 1/4" | against the nook wall, 1 per box |
| Box side (mattress) | 1 | 28 1/8" x 13 1/4" | faces the mattress; hand cut-out to open the lid |
| Box back | 1 | 10 3/4" x 13 1/4" | screw through into wall studs |
| Box front | 1 | 10 3/4" x 13 1/4" | against the back of the front panel; 4 bolts into it |
| Box floor | 1 | 10 3/4" x 25 1/2" | sits 1" off the floor |
| Box lid | 1 | 10 3/4" x 27 5/8" | piano hinge along the long wall-side edge |
| Foot wall liner | 1 | 28 1/8" x 14" | screw to the foot wall studs; runs into the panel groove |
| Foot block | 1 | 1 1/2" x 7 1/4" | above the mattress; glue + 2 screws to the liner, 2 bolts into the panel |

## Key dimensions

Panels: 2 pieces of 4'x4' (1 x 4'x8' sheet(s) of 3/4" plywood, each cut in half)
Heights from floor: mattress top 5", rail top 14", box lids 14", exit sill 4"
Exit opening: 38" to 54" from the head wall; panel halves join at 44"
Head box: 10 3/4" wide outside, 9 1/4" inside, 27 3/4" deep; no foot box (mattress against a liner on the foot wall)

## Hardware

- Wood glue (PVA)
- 1-1/4" wood screws x 14 (box joinery + foot block)
- 1/4-20 threaded inserts for wood x 6, no longer than 3/8" (front panel)
- 1/4-20 x 1" bolts + washers x 6 (box fronts and foot block to front panel)
- 2-1/2" screws x 5 into studs (box back and foot liner)
- Piano hinge 1-1/2" open width, 1 x 24" length(s) (1 per lid)
- Soft-close lid stays x 1 (light-duty toy-box type)
- Shims as needed

## Assembly

1. Cut all parts (sheets/panel_*.dxf). The two front-panel pieces are mirrored in the files: cut them BACK FACE UP so the POCKET layers (box-side grooves 3/8" deep, insert holes 1/2" deep) land on the back. Profile CUT_INSIDE before CUT_OUTSIDE, use tabs/onion skin on small parts, and dry-fit the puzzle joint and one groove on scrap first; adjust knob_clearance / tab_clearance if needed.
2. Ease every exposed edge with a 1/4" roundover (rail top, slots, exit opening, lid edges and the